WORKED SOLUTION

STEP III 1998, Question 6

Pure • Volumes of Polyhedra

STEP III 1998, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s3-q6

The key idea

Put the cube on coordinate axes, so that its vertices are the points whose coordinates are all 0 or a. Distances then come from Pythagoras in three dimensions, and the awkward solids can be measured by cutting simple pieces off the cube. The regular tetrahedron is what is left of a cube once four corners are sliced off, and the regular octahedron is two square-based pyramids placed base to base.

Solution

Part (i)

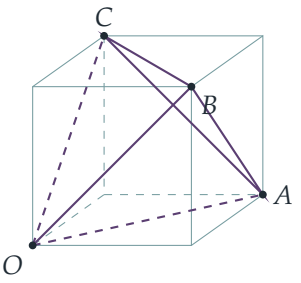
Take the cube $[0,a]^3$. Its vertices are the eight points (x,y,z) with each coordinate 0 or a, and two vertices are adjacent (joined by an edge) exactly when they differ in one coordinate.

Which four vertices can we choose? Call a vertex *even* or *odd* according to how many of its coordinates equal a. Moving along an edge changes one coordinate, so it always swaps even and odd. Suppose our four vertices included an even vertex P and an odd vertex Q. They aren't adjacent, so they differ in three coordinates: Q is the vertex opposite P. Every other vertex is adjacent to P or to Q (the three odd ones to P, the three even ones to Q), so no third vertex could be added. Hence the four vertices are all even,

$$O=(0,0,0),\quad A=(a,a,0),\quad B=(a,0,a),\quad C=(0,a,a),$$

or all odd (the reflection of this set in the plane $x=\frac{1}{2}a$, which gives a congruent solid).

Regularity. Any two of O,A,B,C differ in exactly two coordinates, each by a, so every one of the six distances is $\sqrt{a^2+a^2}=\sqrt{2}\,a$. Each of the four faces is therefore an equilateral triangle of side $\sqrt{2}\,a$, and $OABC$ is a regular tetrahedron. Each of its edges is a diagonal of one face of the cube.



Volume. Removing the tetrahedron from the cube leaves four corner pieces, one at each odd vertex. For instance, the corner at $(a,0,0)$ is the tetrahedron with that vertex and its three neighbours O, A, B. Its three edges at $(a,0,0)$ are mutually perpendicular and of length a, so it is a pyramid with a right-angled triangular base of area $\frac{1}{2}a^2$ and height a, of volume $\frac{1}{3}\cdot\frac{1}{2}a^2\cdot a=\frac{1}{6}a^3$. Hence

$$V_{\text{tet}}=a^3-4\cdot\tfrac{1}{6}a^3=\tfrac{1}{3}a^3.$$

For unit edges we need $\sqrt{2}\,a=1$, that is $a=1/\sqrt{2}$ and $a^3=1/(2\sqrt{2})$. So a regular tetrahedron of unit edge has volume

$$\frac{1}{3}\cdot\frac{1}{2\sqrt{2}}=\frac{1}{6\sqrt{2}}=\frac{\sqrt{2}}{12}.$$

Ask yourself Why do the four corner pieces and the tetrahedron fill the cube exactly, with no overlaps?

Each face of the tetrahedron, such as OAB, cuts off exactly one vertex of the cube, here $(a,0,0)$: the plane $x-y-z=0$ through O, A, B has $(a,0,0)$ strictly on one side and the other seven vertices on the other side or on it. So the cube splits into the four corner pieces, one beyond each face, and the tetrahedron in the middle.

Part (ii)

A regular octahedron of unit edge can be placed with its six vertices at

$$(\pm h,0,0),\quad (0,\pm h,0),\quad (0,0,\pm h),\qquad h=\tfrac{1}{\sqrt{2}},$$

where each vertex is joined to the four that are not opposite it. Two such vertices, for example $(h,0,0)$ and $(0,h,0)$, are at distance $\sqrt{2}\,h=1$. So all twelve edges have length 1, and the eight faces, one in each octant, are equilateral. The four vertices in the plane $z=0$ form a square of side 1. The octahedron is two square-based pyramids on this square, with apexes $(0,0,\pm h)$, each of height h. Hence

$$V_{\text{oct}}=2\cdot\tfrac{1}{3}\cdot 1^2\cdot\frac{1}{\sqrt{2}}=\frac{\sqrt{2}}{3}.$$

More generally, scaling lengths by e scales volumes by e^3, so a regular octahedron of edge e has volume $\frac{\sqrt{2}}{3}e^3$.

Part (iii)

It is now neater to centre the cube of side a at the origin, as $[-\frac{1}{2}a, \frac{1}{2}a]^3$. Its face centres are

$$(\pm\tfrac{1}{2}a, 0, 0), \quad (0, \pm\tfrac{1}{2}a, 0), \quad (0, 0, \pm\tfrac{1}{2}a).$$

This is exactly the configuration of part (ii) with $h = \frac{1}{2}a$. The centres of opposite faces are at distance a. The centres of adjacent faces are at distance $\sqrt{(\frac{1}{2}a)^2 + (\frac{1}{2}a)^2} = a/\sqrt{2}$. The convex solid with these six vertices has eight faces, one in each octant: for example, the face in the octant $x, y, z \geq 0$ is the triangle with vertices $(\frac{1}{2}a, 0, 0)$, $(0, \frac{1}{2}a, 0)$, $(0, 0, \frac{1}{2}a)$. Each face is equilateral with side $a/\sqrt{2}$, so the face centres form a regular octahedron of edge $e = a/\sqrt{2}$.

By part (ii), its volume is

$$\frac{\sqrt{2}}{3}\left(\frac{a}{\sqrt{2}}\right)^3 = \frac{\sqrt{2}}{3} \cdot \frac{a^3}{2\sqrt{2}} = \frac{a^3}{6}.$$

The tetrahedron of part (i), whose vertices are four of the vertices of the same cube, has volume $\frac{1}{3}a^3$. Since $\frac{1}{6}a^3 = \frac{1}{2} \cdot \frac{1}{3}a^3$, the octahedron has half the volume of the tetrahedron, as required.

Checks

- Using the base-times-height formula with base area $\frac{\sqrt{3}}{4}$ and height $\sqrt{2/3}$ gives the unit tetrahedron volume $\frac{1}{3} \cdot \frac{\sqrt{3}}{4} \cdot \sqrt{\frac{2}{3}} = \frac{\sqrt{2}}{12} \approx 0.11785$, which agrees with part (i).
- For equal edges, $V_{\text{oct}} = 4V_{\text{tet}}$, since $\frac{\sqrt{2}}{3} = 4 \cdot \frac{\sqrt{2}}{12}$. This matches the classical dissection of a tetrahedron of edge 2 into four tetrahedra of edge 1 (at its corners) and one octahedron of edge 1 (in the middle): measured in unit tetrahedra, $8 = 4 + 4$.
- Numerically, a determinant computation over all choices of four non-adjacent cube vertices gave volume $\frac{1}{3}$ for the unit cube, and a Monte Carlo estimate of the region $|x| + |y| + |z| \leq 1/\sqrt{2}$ gave 0.4715, against $\sqrt{2}/3 \approx 0.47140$.

Common slips

- Taking the cube's side to be 1 in part (i): the tetrahedron's edge is then $\sqrt{2}$, not 1, and its volume has to be rescaled by $(1/\sqrt{2})^3$.
- Subtracting the wrong corners: there are four corner pieces (one at each vertex not used), each of volume $\frac{1}{6}a^3$, not eight.
- In part (iii), using the cube's side as the octahedron's edge. Adjacent face centres are $a/\sqrt{2}$ apart.

Beyond STEP

The two tetrahedra inside a cube, and the octahedron of face centres, are the start of the story of the Platonic solids and their symmetry groups: the rotations of the cube that preserve one inscribed tetrahedron form a group of order 12, half of all 24 rotations of the cube. The cube and octahedron are *dual* (face centres of one give the vertices of the other), and volumes of such solids come out neatly from determinants, which measure how a linear map scales volume. Strang's *Introduction to Linear Algebra* treats determinants as volumes, and Armstrong's *Groups and Symmetry* studies the symmetry groups of these solids.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.